An example is the interior of black holes. Another is what happened to the universe during the Big Bang. In more general terms, we do not know how time and space behave at very small scale. In all these instances, today's theories become confused and no longer tell us anything reasonable: quantum mechanics cannot deal with the curvature of spacetime, and general relativity cannot account for quanta. This is the problem of quantum gravity.

The problem goes even deeper. Einstein understood that space and time are manifestations of a physical field: the gravitational field. Bohr, Heisenberg and Dirac understood that physical fields have a quantum character: granular, probabilistic, manifesting through interactions. It follows that space and time must also be quantum entities possessing these strange properties.

What, then, is quantum space? What is quantum time? This is the problem we call quantum gravity. A band of theoretical physicists scattered across five continents is laboriously seeking to solve the problem. Their objective is to find a theory, that is to say, a set of equations – but, above all, a coherent vision of the world – with which to resolve the current schizophrenia between quanta and gravity.

It isn't the first time that physics has found itself faced with two highly successful but apparently contradictory theories. The effort to synthesize has in the past been rewarded with great strides forward in our understanding of the world. Newton discovered universal gravity precisely by combining Galileo's physics of how things move on Earth with Kepler's physics of the heavens. Maxwell and Faraday found the equations of electromagnetism by bringing together what was known about electricity and what was known about magnetism. Einstein found special relativity in order to resolve the apparent conflict between Newton's mechanics and Maxwell's electromagnetism – and then general relativity in order to resolve the resulting conflict between Newton's mechanics and his own special relativity.

Theoretical physicists are thus only too happy when they discover a conflict of this type: it is an extraordinary opportunity. The question to ask is: can we construct a conceptual structure compatible with what we have learned about the world with *both* theories?

To comprehend what quantum space and quantum time are, we need once more to revise in depth the way we conceive things. We need to rethink the grammar of our understanding of the world. Just as happened with Anaximander, who understood that Earth flies in space, and that 'up' and 'down' do not exist in the cosmos; or with Copernicus, who understood that we are moving across the heavens at great speed; or with Einstein, who understood that spacetime squashes like a mollusc, and time passes differently in different places … once again, in seeking a coherent vision of the world in keeping with what we have learned about it, our ideas about the nature of reality have to change.